

Micro-Doppler Feature Extraction Based on Time-Frequency Spectrogram for Ground Moving Targets Classification With Low-Resolution Radar

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Abstract—A novel feature extraction method based on micro-Doppler signature is proposed to categorize ground moving targets into three kinds, i.e., single walking person, two people walking, and a moving wheeled vehicle. Signal models and measured data from a low-resolution radar are first analyzed to find the differences between the micro-Doppler signatures from the three kinds of considered targets. Then, such discriminative micro-Doppler signatures are represented by a 3-D feature vector extracted from the time-frequency spectrograms. In the experiments based on the measured data, the ratio of the between-class distance to the within-class distance, which is defined based on Fisher discriminant analysis, is exploited to assess the discriminative ability of the 3-D feature vector. Moreover, support vector machine classifier is utilized to evaluate the classification performance. Experimental results show that the proposed micro-Doppler features can achieve a good discriminative ability and a satisfactory classification performance.

Index Terms—Micro-Doppler, time-frequency analysis, feature extraction, radar target classification.

I. INTRODUCTION

WITH the development of new technology in the radar field and the increasing demand for security applications such as intruder detection and site security monitoring, there is an exigent demand to equip a low resolution radar with target classification capability. However, since most low resolution radars exhibit a range resolution much larger than the target size, a target is usually regarded as point-like, losing the possibility to resolve its structure and some characteristic details. Thus finding practical classification methods for low resolution radar is an important research subject in the field of Radar Automatic Target Recognition (RATR).

It is well known that moving targets will cause a Doppler frequency shift into the radar returned signal, referred to as Doppler Effect [1]. If there are structural components rotating or oscillating on the target, these moving components will

induce additional frequency modulations around the centre frequency of the target, which is referred to as micro-Doppler effect. Micro-Doppler effect is widely spread in the real world, e.g., the swinging arms and legs of a walking human, the precession of space cone-shaped targets and the Jet Engine Modulation (JEM) characteristics, i.e. the effect caused by the rotating fans in the aircraft. Chen. *et al.* introduced the notion of micro-Doppler effect in radar, and provided some detailed mathematical models for the analysis of different kinds of micro-Doppler modulations [2], [3]. Since different kinds of targets usually induce different types of micro-Doppler modulations in the low resolution radar returns, some discriminative features may be extracted from the mentioned modulations to describe the different targets. As a consequence, there have been lots of researches applying the micro-Doppler effect to the radar target detection and classification field [4]–[9]. After analyzing the related signatures for pedestrians and vehicles, [10], [11] extract and classify the corresponding micro-Doppler features. In [12], [13], the micro-Doppler information contained in the time-frequency spectrograms of different human gait motions is utilized for human activity classification. [14] proposes a hierarchical micro-Doppler feature extraction method to categorize moving vehicles into two kinds, i.e., wheeled vehicle and tracked vehicle. References [15]–[18] analyze the JEM characteristics of the airplane targets and investigate the airplane classification problem based on the JEM effects, of which [15], [17] analyze the scattering characteristics of the rotating blades in a helicopter, [16] utilizes the measured JEM characteristics to realize the radar target identification and [18] proposes a new feature extraction method based on Empirical Mode Decomposition (EMD) and CLEAN technique. Compared with the features used with high resolution radar [19]–[22], the micro-Doppler features represent the different micro-motion characteristics from different target categories for target classification (which means to decide the target category), rather than describe the detailed waveforms of the training echoes from some given target models for target recognition (which means to decide the detailed target model), to guarantee the generalization performance among different target models from the same target category.

The main innovation of the present paper is the definition of a novel 3-dimensional micro-Doppler feature vector extracted

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from the time-frequency spectrograms to represent the micro-motion characteristics of single walking person, two people walking and a moving wheeled vehicle. The ratio of the between-class distance to the within-class distance, which is defined based on Fisher Discriminant Analysis (FDA) [23], shows that the proposed features have a higher degree of discriminative ability than some existing features. Additionally, the resulting average classification accuracy obtained via a Support Vector Machine (SVM) [24] classifier is about 96% on measured data.

This paper is organized as follows. In Section II, we analyze the signal models and measured data from a low resolution radar to find the differences between the micro-Doppler signatures from the three kinds of considered targets. In Section III, we propose the novel feature extraction method based on the time-frequency spectrograms of measured data to depict the different micro-Doppler characteristics of the considered targets. In Section IV, the ratio of the between-class distance to the within-class distance and SVM classifier are exploited to evaluate the performance of our proposed features. Finally, we give concluding remarks in Section V.

II. MICRO-DOPPLER SIGNATURES OF MEASURED DATA

A. Analytic Models for Walking Person and Moving Wheeled Vehicle

When a person walks, there are various types of simultaneous movements such as the swinging motions of the arms and legs, the lateral translation motion of the spine, the flexing motion of the hip, the rotation of the shoulders, et al. Since different kinds of movements need different representations, the mathematical model for a walking person is too complex for an analytic description. However, since all the simultaneous movements in a person walking are periodic, they induce periodic micro-Doppler modulations in the radar return. The instantaneous frequency of each periodic modulation may be approximately formulated as $A_k \sin(\omega_k t + \varphi_k)$, where φ_k denotes the initial phase of the k th movement, A_k and ω_k denote the amplitude and the frequency of the k th movement respectively. We empirically know that A_k and ω_k are related to the motion radius and the walking speed.

Since the lengths of arms and legs are much larger than the motion radii of the lateral translation of the spine and the rotation of the shoulders, the micro-Doppler modulations induced by the swinging arms and legs will capture a much wider frequency bandwidth than those induced by the lateral translation of the spine and the rotation of the shoulders. The latter are regarded as the modulations induced by torso components.

As to the micro-Doppler modulations induced by the rolling wheels of a moving wheeled vehicle, they are quite different from those of a walking person. Assume there are N scattering points distributed evenly on the wheeled edge and the initial phase of the reference scattering point is θ_0 . As analyzed in [14], the radar returned micro-Doppler signal of the rolling wheel can be represented as:

$$s(t) = \sum_{n=1}^N \zeta_n \sum_{m=-\infty}^{+\infty} \exp(jm\theta_n) J_m(\beta) \exp(jm\omega t), \quad (1)$$

where ζ_n is the scattering coefficient of the n th scattering point, ω is the rotation radian frequency of the rolling wheel, $\beta = 4\pi r_0/\lambda$ with r_0 denoting the rotation radius and λ being the wavelength of the transmitted signal, $J_m(\cdot)$ is the Bessel function of the first kind and order m , $\theta_n = \theta_0 + \frac{2\pi n}{N}$ with θ_n being the initial phase of the n th scattering point. In addition, when a wheeled vehicle is moving, the bulk component can be regarded as an ensemble due to the rigid structure characteristics of a vehicle. Therefore, the modulation induced by the bulk component should be represented as $A \exp(j\frac{4\pi}{\lambda} vt)$, where v denotes the speed of the translation motion and A denotes the scattering coefficient of the bulk component.

Moreover, since the material of a wheel is rubber which hardly scatters the electromagnetic wave, the micro-Doppler modulation components induced by the rolling wheel have a simple structure which mainly spread within the frequency range $[0, \frac{2v}{\lambda}]$. The Doppler modulation induced by the bulk component is a single frequency and it captures most of the energy of the radar returns from a moving wheeled vehicle.

B. Analysis of Micro-Doppler Signatures of Measured Data

Our measured data used in this paper are collected from the three kinds of targets, i.e., single walking person, two people walking and a moving wheeled vehicle. As analyzed above, if the two people walk in a line with their projected velocities onto the radar line-of-sight (LOS) and all their micro-motions, including the swinging motions of the arms and legs, the lateral translation motion of the spine, the flexing motion of the hip, the rotation of the shoulders, et al, are exactly same, their micro-Doppler signature in the time-frequency spectrogram should be very similar to that of single walking person. To discriminate single walking person and two people walking, this case doesn't happen in our experiment.

The low resolution radar used in the experiment is a Ka band system with its Pulse Repetition Frequency (PRF) F_{PRF} being 3KHz, its dwell time being 1.2s, the pulse length being 3 us for each transmitted Linear Frequency Modulated (LFM) pulse and the bandwidth being 18MHz. Since the range resolution is about 8.3 m, there are range migrations between the neighbouring range cells during the 1.2s dwell time for the moving wheeled vehicle in our experiment, which is of about 5 m in length. Thus the matched filter and Moving Target Detection (MTD) are first utilized for some continuous pulses along the range dimension to detect the range cell where the target situates. Then we can get a slow time acquisition consisting of 3600 samples via the sampling processing from the range cells where the target situates with the sampling rate of PRF during the 1.2s dwell time for each target measurement. We have totally 120 acquisitions from single person that walks towards the radar and totally 120 acquisitions from two people walking towards or away from the radar, whose walking speeds are close but not the same. There are totally 120 acquisitions from a moving wheeled vehicle which recedes or approaches the radar.

In our original measured data, the target signature and ground clutter are mixed in the returned echo of a ground

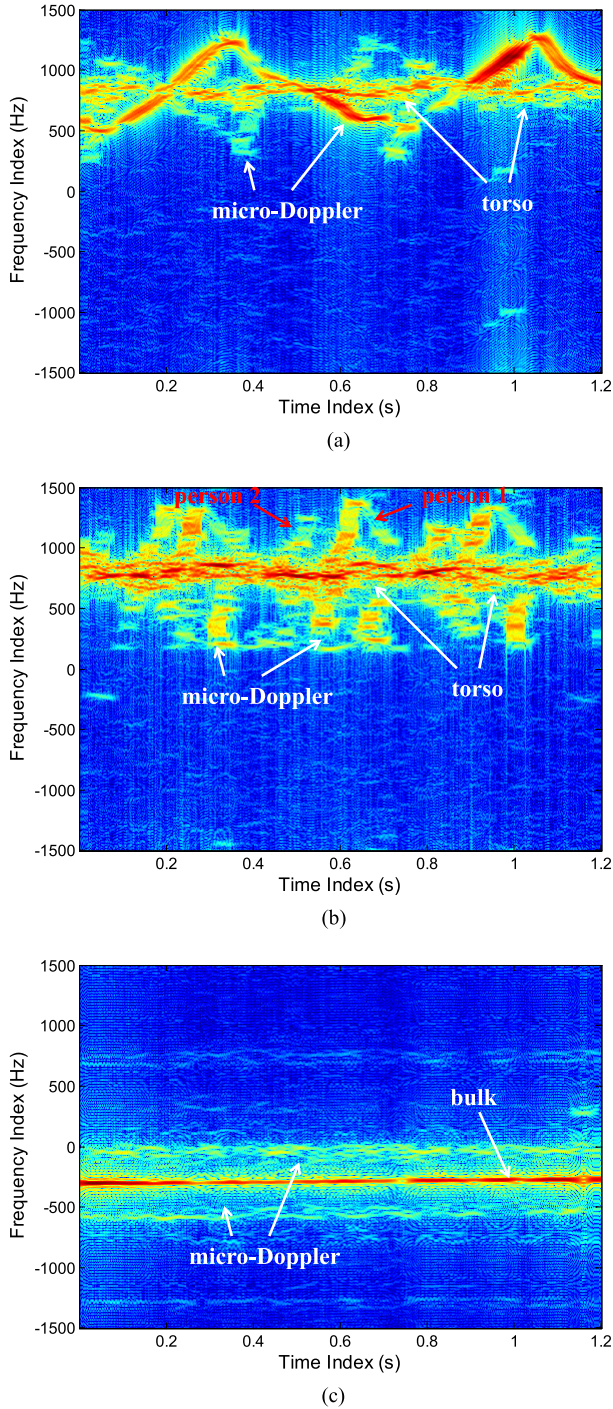


Fig. 1. Time-frequency spectrograms of some measured data: (a) single walking person; (b) two people walking; and (c) a moving wheeled vehicle. Here the ground clutter components in the original data have been removed via the CLEAN technique [25].

moving target. Here the Doppler spectrum of ground clutter component mainly corresponds to the frequency components around the frequency index $f = 0$. Before micro-Doppler feature extraction, we use the CLEAN technique [25], a traditional ground clutter suppression method, to remove the ground clutter components in our original measured data. Figures 1 (a), (b) and (c) show the time-frequency spectrograms of the ground clutter removed measured data from single walking person, two people walking and a moving wheeled

vehicle respectively. Each time-frequency spectrogram is the modulus matrix of Short Time Fourier Transform (STFT) of the acquisition. A 43ms Hamming window is used in the STFT, then the frequency resolution is about 23.26Hz. In Figure 1, the frequency range in the time-frequency spectrograms depends on the PRF of the radar system. In our experiments, since the targets, single person, two people and wheeled vehicle, move towards or away from the radar, their Doppler frequencies can be positive or negative. Therefore, the frequency range is $\pm \frac{F_{PRF}}{2} = \pm 1.5\text{KHz}$. In addition, in Figure 1, we indicate the micro-motion components, i.e., those from the swinging arms or legs of walking person (people) or the rolling wheels of a moving vehicle, and the translation-motion components, i.e. those from the torso of the walking person (people) or the bulk of a moving vehicle.

The time-frequency spectrograms of measured data depicted in Figure 1 are in accordance with the above analysis, i.e., the instantaneous micro-Doppler frequency varies periodically with time for walking person (people) while that of a moving vehicle does not show such periodic behavior. In addition, the frequency modulations induced by the torso component are also time variant though the frequency range is much smaller than the modulations induced by the swinging arms and legs. However, the frequency modulation induced by the bulk motion of a moving wheeled vehicle captures a single Doppler bin in the time-frequency spectrograms as shown in Figure 1(c). From the energy distribution of the time-frequency spectrograms we can see that for the moving vehicle situation depicted in Figure 1(c), the energy of the translation motion is much larger than that of the micro-Doppler component. On the other hand, for the walking person (people) case depicted in Figure 1 (a) and (b), the energy of the micro-Doppler component is close to that of the translation motion component and under some situations it may be larger than that of the translation motion component.

III. TIME-FREQUENCY FEATURE EXTRACTION

Figure 1 highlights that there are some differences among the micro-Doppler modulations of the considered targets. We use the following three features to depict the differences of the micro-Doppler signatures reflected in the time-frequency spectrograms of the considered targets.

A. Symbol Notation

1) $\tilde{\mathbf{S}} = [\tilde{S}_{f,t}]_{f=1,t=1}^{N_F,N_T}$ denotes the time-frequency spectrogram, which is the modulus matrix of STFT of the acquisition. Here f is the row index corresponding to the frequency dimension in STFT, t is the column index corresponding to the time dimension in STFT, N_F represents the length of the Fast Discrete Fourier Transformation (FDFT) for each short time signal obtained by the Hamming window with the length of $\tau = 43\text{ms}$, and N_T represents the number of samples in the time dimension in STFT. In our experiments, we use the FDFT with the length of 3000 and add two zero sequences with the lengths of $n = \frac{\tau \cdot F_{PRF} - 1}{2}$ to the two ends of the slow time acquisition to calculate the STFT for each acquisition, therefore, $N_F = 3000$ and $N_T = T \cdot F_{PRF} = 3600$, i.e., the size of matrix $\tilde{\mathbf{S}}$ is 3000×3600 . $\bar{\mathbf{S}} = (1/N_F \times N_T)$

$\sum_{f=1}^{N_F} \sum_{t=1}^{N_T} \tilde{S}_{f,t}$ denotes the mean of the time-frequency spectrogram.

2) $\mathbf{E} = [E_f]_{f=1}^{N_F} = [\frac{1}{N_T} \sum_{t=1}^{N_T} S_{f,t}]_{f=1}^{N_F} = \frac{1}{N_T} [\sum_{t=1}^{N_T} S_{f,t}]_{f=1}^{N_F}$ denotes the mean vector consisting of mean value of each row in the normalized time-frequency spectrogram with $S_{f,t} = (\tilde{S}_{f,t} / \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} \tilde{S}_{f,t})$; since $\sum_{f=1}^{N_F} \sum_{t=1}^{N_T} S_{f,t} = 1$, $\bar{E} = \frac{1}{N_F} \sum_{f=1}^{N_F} E_f = \frac{1}{N_F} \frac{1}{N_T} \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} S_{f,t} = \frac{1}{N_F \cdot N_T}$ denotes the mean of $\mathbf{E}$; $f_{\max} = \arg \max \mathbf{E}$ denotes the index corresponding to the maximum value in $\mathbf{E}$; f_{upper} denotes the index that satisfies $E_{f_{\text{upper}}} \geq \bar{E}$, $E_{f_{\text{upper}}+1} < \bar{E}$ and nearest to $f_{\max}$ within the range $[f_{\max}, N_F]$; f_{lower} denotes the index that satisfies $E_{f_{\text{lower}}} \geq \bar{E}$, $E_{f_{\text{lower}}-1} < \bar{E}$ and nearest to $f_{\max}$ within the range $[1, f_{\max}]$. The above definitions are illustrated in the left subfigures in Figure 2.

3) $\mathbf{G} = [G_t]_{t=1}^{N_T}$ with $G_t = \arg \max_f ([\tilde{S}_{f,t}]_{f=1}^{N_F})$, where G_t denotes the row index of the largest value in the t th column of $\tilde{S}$; $\bar{G} = \frac{1}{N_T} \sum_{t=1}^{N_T} G_t$ denotes the mean of $\mathbf{G}$. The above definitions are illustrated in the bottom subfigures in Figure 2.

We give the feature extraction examples for the three kinds of considered targets in Figure 2. In each example, the time-frequency spectrogram of the measured data, i.e., $\tilde{S}$, is shown in the middle subfigure, the mean vector consisting of mean value of each row in the normalized time-frequency spectrogram, i.e., $\mathbf{E}$, is shown in the left subfigure, the frequencies corresponding to the largest values in each column of the time-frequency spectrogram, i.e., $\mathbf{G}$, is displayed in the bottom subfigure. The measured data used in Figure 2 are the same as those in Figure 1.

Feature 1: the variance of the time-frequency spectrogram.

$$\text{Feature}(1) = \frac{1}{N_F \times N_T} \sum_{f=1}^{N_F} \sum_{t=1}^{N_T} (\tilde{S}_{f,t} - \bar{S})^2, \quad (2)$$

Variance is a quantity that describes the fluctuation characteristic of a signal. Severely fluctuating signals induce a large variance. In other words, the value of *Feature*(1) is determined by the fluctuation degree of the time-frequency plane. For the time-frequency spectrogram of a moving vehicle shown in the middle subfigure of Figure 2(c), the whole time-frequency plane is smooth beside the range captured by the bulk component. Thus the corresponding value of *Feature*(1) of a moving wheeled vehicle is the largest. Since there are more abundant micro-Doppler modulations in the time-frequency spectrograms of two people walking than that of single walking person, the value of *Feature*(1) of two people walking is smaller than that of single walking person. The above analysis is in accordance with the results shown in Table I, i.e., the time-frequency spectrogram of a moving vehicle has the largest variance of the three kinds of target while the time-frequency spectrogram of two people walking has the smallest variance.

Feature 2: the bandwidth of the targets' Doppler modulations in the time-frequency spectrogram.

$$\text{Feature}(2) = f_{\text{upper}} - f_{\text{lower}}, \quad (3)$$

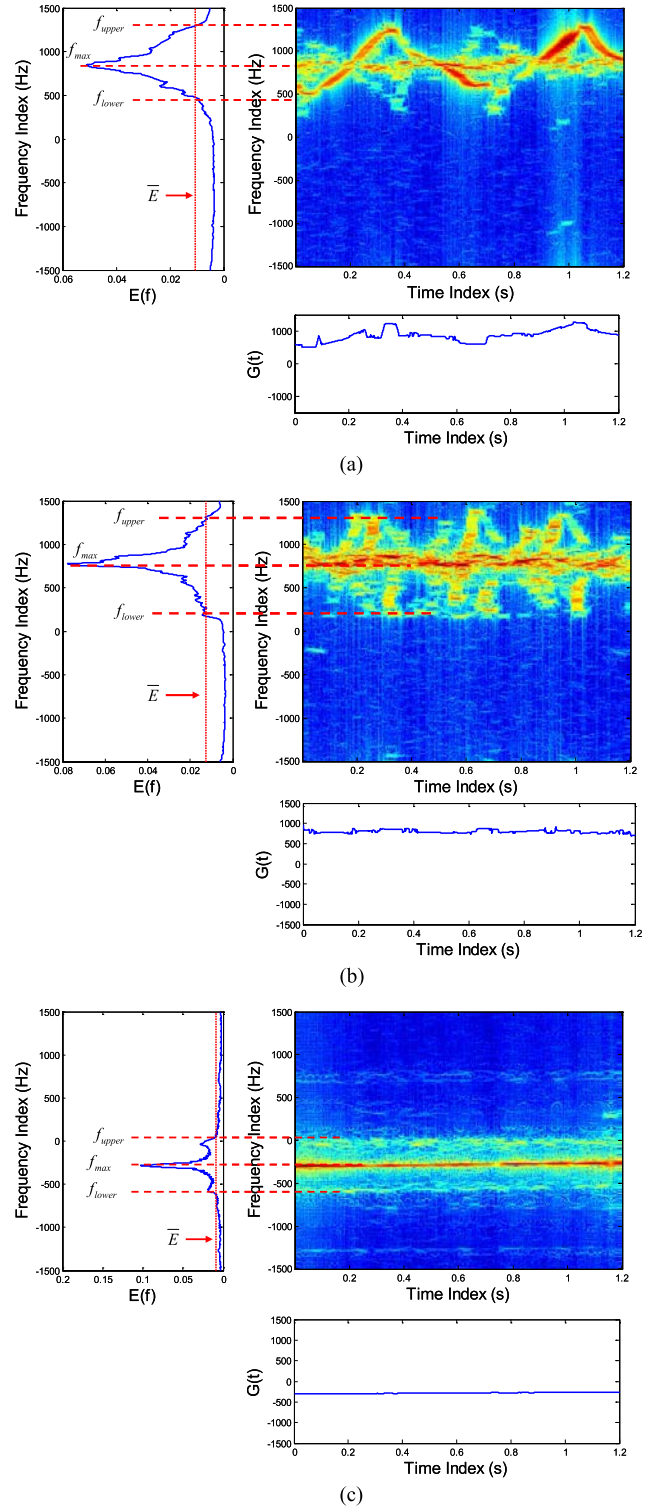


Fig. 2. Feature extraction examples for the measured data from the three kinds of considered targets, where the time-frequency spectrogram, i.e., $\tilde{S}$, is shown in the middle subfigure, the mean vector consisting of mean value of each row in the normalized time-frequency spectrogram, i.e., $\mathbf{E}$, is shown in the left subfigure, and the frequencies corresponding to the largest values in each column of the time-frequency spectrogram, i.e., $\mathbf{G}$, is shown in the bottom subfigure: (a) single walking person; (b) two people walking; and (c) a moving wheeled vehicle.

As shown in the left subfigures in Figure 2, the bandwidth of the targets' Doppler modulations for the single walking person is much wider than that of a moving vehicle; while two people

TABLE I
AVERAGED FEATURE VALUES OF THE 120 ACQUISITIONS FOR EACH KIND OF CONSIDERED TARGET

	<i>Feature(1)</i>	<i>Feature(2)</i>	<i>Feature(3)</i>
single walking person	1.7109×10^{-5}	750.2667	82.4420
two people walking	5.9909×10^{-6}	986.3333	42.1851
moving wheeled vehicle	2.7462×10^{-5}	612.1000	17.2689

walking have a wider induced modulation bandwidth than a single walking person due to more complicated micro-Doppler modulation. The above analysis is in accordance with the values of *Feature(2)* shown in Table I.

Feature 3: the variance of the frequencies corresponding to the largest values in each column of the time-frequency spectrogram.

$$Feature(3) = \frac{1}{N_T} \sum_{t=1}^{N_T} (G_t - \bar{G})^2, \quad (4)$$

Since the energy of the bulk motion is much larger than that of the micro-Doppler motion for a moving vehicle, the instantaneous frequencies corresponding to the largest values in each column of the time-frequency spectrogram always correspond to the bulk motion component. As shown in the bottom subfigure of Figure 2(c), when a vehicle is moving, the corresponding frequency induced by the moving bulk is close to a constant within the short dwell time. However, as discussed in Section II, the torso motion consists of the lateral translation motion of the spine, the flexing motion of the hip, the rotation of the shoulders, et al. Then the frequency modulation induced by the torso motion corresponds to a much wider frequency range than that induced by the bulk motion, and the energy of the torso motion is more spread than that of the bulk motion as illustrated in the middle subfigures of Figure 2. From the bottom subfigures of Figure 2(a) and (b) we can see that the curves of G for single walking person and two people walking both vary with time, therefore, their values of *Feature(3)* are much larger than that of a moving vehicle. For a single walking person, the energy of the micro-Doppler component is close to that of the torso component. However, for two people walking, the energy of the torso component is often larger than that of the micro-Doppler component since the arms and legs are easily sheltered. As shown in the bottom subfigures of Figure 2(a) and (b), the curve of G for single walking person fluctuates more severely than that of two people walking. Thus the value of this feature for single walking person is larger than that of two people walking. The results in Table I confirm the above analysis.

IV. MICRO-DOPPLER SIGNATURES OF MEASURED DATA

In this section, we utilize the measured data to evaluate the classification performance of the proposed 3-dimensional feature vector. First, we divide the measured data into two sets, i.e., training data set and test data set. For each kind of target, we use half of the measured data as the training data set, i.e. 60 acquisitions from single walking person, 60 acquisitions from two people walking and 60 acquisitions from a moving

wheeled vehicle. The rest of measured data are utilized as the test data set. Then the 3-dimensional micro-Doppler feature is extracted from the time-frequency spectrogram of each acquisition in the training data set and test data set. In the training stage, we use the feature data of training acquisitions to train a SVM classifier; while in the test stage the trained classifier is utilized to categorize each feature data from test acquisitions into corresponding category. In addition, before putting the feature data into a classifier, we need to normalize them via subtracting their mean and divide their standard deviation to unify the scale of different features.

A. Micro-Doppler Feature Analysis

In order to evaluate the discriminative ability of the proposed 3-dimensional feature vector, we utilize the ratio of the between-class distance to the within-class distance, which is defined based on FDA [23]. An ideal feature vector will achieve the large between-class distance and small within-class distance. The within-class distance is defined as [23]:

$$S_W = \sum_{k=1}^K S W_k^2, \quad (5)$$

Where $S W_k = \sum_{x \in C_k} (x - m_k)^2$, $x \in C_k$ denotes the feature vector corresponding to the k th class C_k , m_k is the mean vector of the k th class, i.e., $m_k = \frac{1}{N_k} \sum_{x \in C_k} x$ with N_k denoting the number of vectors in the k th class. The small within-class distance means the feature vectors are close to each others in the category.

The between-class distance is defined as:

$$S_B = \sum_{k=1}^K \sum_{j>k}^K (m_k - m_j)^2, \quad (6)$$

Large between-class distance means that the class mean of each category is far away from others.

Table II gives the values of between-class distance, within-class distance and the ratio of the between-class distance to the within-class distance for the feature vectors obtained from the training set (for each training feature, the corresponding category is known). In Table II, *Feature(1)*, *Feature(2)* and *Feature(3)* are the proposed 3-dimensional feature described in Section III. We also calculate the between-class distance and within-class distance for the 3-dimensional feature vector described in paper [11] and the 4-dimensional feature vector described in paper [13] for comparison. In Table II, *Feature(4)*, *Feature(5)* and *Feature(6)* are the 3-dimensional feature vector described in paper [11]; while *Feature(7)*,

TABLE II

VALUES OF WITHIN-CLASS DISTANCE, BETWEEN-CLASS DISTANCE AND THE RATIO OF THE BETWEEN-CLASS DISTANCE TO THE WITHIN-CLASS DISTANCE FOR DIFFERENT FEATURES. *Feature(1) ~ Feature(3)* ARE PROPOSED IN SECTION III, *Feature(4) ~ Feature(6)* ARE PROPOSED IN PAPER [11] AND *Feature(7) ~ Feature(10)* ARE PROPOSED IN PAPER [13]

	S_W	S_B	S_B/S_W
<i>Feature(1)</i>	1917.7	5.9379	3.097×10^{-3}
<i>Feature(2)</i>	3153.7	5.0948	1.616×10^{-3}
<i>Feature(3)</i>	1003.0	6.6270	6.605×10^{-3}
<i>Feature(4)</i> ^[11]	10094.1	2.1884	2.168×10^{-4}
<i>Feature(5)</i> ^[11]	9166.0	1.4049	1.533×10^{-4}
<i>Feature(6)</i> ^[11]	7128.8	2.0520	2.878×10^{-4}
<i>Feature(7)</i> ^[13]	4.7631×10^{13}	7.3944×10^4	1.5524×10^{-9}
<i>Feature(8)</i> ^[13]	6.9422×10^{14}	8.8989×10^5	1.2819×10^{-9}
<i>Feature(9)</i> ^[13]	1.4417×10^{16}	9.8798×10^5	6.8530×10^{-11}
<i>Feature(10)</i> ^[13]	5.3095×10^{13}	3.9234×10^5	7.3894×10^{-9}

TABLE III

CONFUSION MATRIX OF THE CLASSIFICATION ACCURACIES FOR EACH CONSIDERED TARGET CATEGORY AND THE AVERAGE CLASSIFICATION RATE VIA THE PROPOSED 3-DIMENSIONAL FEATURE VECTOR (%)

	Single	Two	Vehicle	Average
Single	91.67	1.670	1.670	96.11
Two	1.670	98.33	0	
Vehicle	6.670	0	98.33	

TABLE IV

CONFUSION MATRIX OF THE CLASSIFICATION ACCURACIES FOR EACH CONSIDERED TARGET CATEGORY AND THE AVERAGE CLASSIFICATION RATE VIA THE 3-DIMENSIONAL FEATURE VECTOR PROPOSED IN [11] (%)

	Single	Two	Vehicle	Average
Single	80.00	3.330	11.67	87.78
Two	8.330	95.00	0	
Vehicle	11.67	1.670	88.33	

Feature(8), *Feature(9)* and *Feature(10)* are the 4-dimensional feature vector described in paper [13]. Based on the results in Table II, we can see that our proposed features achieve the smaller within-class distance and larger between-class distance than *Feature(4) ~ Feature(10)*, i.e., in the feature domain, the proposed *Feature(1)*, *Feature(2)* and *Feature(3)* are more discriminative than *Feature(4) ~ Feature(10)*.

B. Classification Results

The features extracted in the training stage are utilized to train a SVM which is then used to determine the category of each test acquisition. The SVM is fundamentally a two-class classifier. There is no uniform method for combining multiple two-class SVMs in order to build a multi-class classifier. In the following, we use the quadratic-programming based SVM (QP-SVM) proposed in [24]. This classifier defines a single objective function for training all one-versus-the-rest two-class SVMs simultaneously, based on maximizing the margin for each two-class SVM. Contrast to the one-versus-the-rest (OAA) [26] and one-versus-one (OAO) [27]

approaches, the QP-SVM can avoid ambiguities in the resulting classification. The main disadvantage of the QP-SVM is the much slower training for a large number of categories. Nevertheless, here we only consider the three-class classification problem and the computation time is reasonable. In our experiment, we choose the Gaussian function as the kernel of SVM, the parameter of Gaussian kernel is tuned in the range [0, 5] and we select the best classification result for the following analysis and comparison.

Tables III, IV and V show the confusion matrixes of the classification accuracies for each target category and the average classification rate via the proposed 3-dimensional feature vector described in Section III, the 3-dimensional feature vector of [11] and the 4-dimensional feature vector proposed in [13] respectively. The average classification rate is calculated using the number of test acquisitions categorized into its right class divided by the totally number of test acquisition. From the results shown in Table III, we can see that our proposed method can categorize test samples of the three kinds of considered targets into the right class with the accuracy over 90%. However, as shown in Table IV,

TABLE V
CONFUSION MATRIX OF THE CLASSIFICATION ACCURACIES FOR EACH CONSIDERED TARGET CATEGORY AND THE AVERAGE CLASSIFICATION RATE VIA THE 4-DIMENSIONAL FEATURE VECTOR PROPOSED IN [13] (%)

	Single	Two	Vehicle	Average
Single	65.00	3.330	0	84.44
Two	5.00	88.33	0	
Vehicle	30.00	8.340	100	

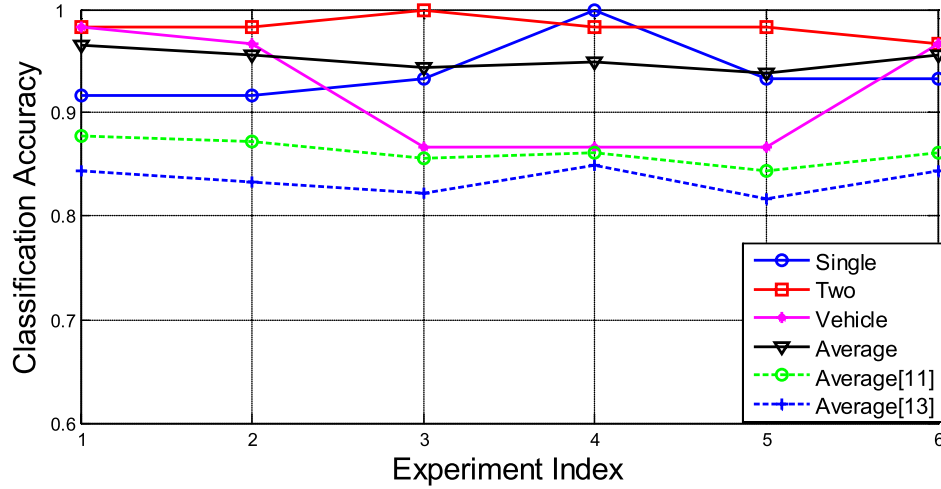


Fig. 3. The classification accuracies from the different partitions of the training and test sets. The four real lines show the results of each considered target category and their average values via the proposed 3-dimensional feature vector; while the two dashed lines show the average classification rates via the 3-dimensional feature vector proposed in [11] and the 4-dimensional feature vector proposed in [13].

the 3-dimensional feature vector proposed in [11] wrongly recognize the test acquisitions of the single walking person to be those of the moving vehicle with the rate over 10%, and vice versa. As shown in Table V, the 4-dimensional feature vector proposed in [13] tends to wrongly recognize the test acquisitions of the single walking person to be those of the moving vehicle with the rate over 30%; and it tends to wrongly recognize the test acquisitions of the two people walking to be those of the moving vehicle with the rate over 8%. Therefore, comparing the results in Tables III, IV and V, our proposed 3-dimensional feature vector achieves a much higher classification accuracy than the 3-dimensional feature vector of [11] and the 4-dimensional feature vector of [13].

In order to evaluate the influence of different partition criteria of the training set and test set on the classification performance of our proposed 3-dimensional feature vector and two existing feature vectors proposed in [11] and [13], we repeat the classification experiment six times with different partitions of the training and test sets. Specifically, for each classification experiment in the following, we randomly choose half of the acquisitions of each target as the training set, the rest of the measured data as the test set. Figure 3 shows the classification accuracies of each considered target category for our proposed feature vector and the average classification rates for the three feature vectors from the different experiments. From Figure 3, we can see that the classification performance of our proposed 3-dimensional feature vector is not only better than the existing feature vectors proposed in [11] and [13], but also robust with respect to partitions of the training set and test set.

V. CONCLUSION

Micro-Doppler signatures induced by the rotating or oscillating parts on the target can be utilized as the unique features to categorize radar targets into different kinds. In this paper, we propose a novel feature extraction method based on the micro-Doppler signatures in the time-frequency spectrograms to categorize ground moving targets into three kinds, i.e., single walking person, two people walking and a moving wheeled vehicle. Experiment results based on measured data show the proposed micro-Doppler features have a higher discriminative ability and higher classification accuracy than some existing features.

Here we only discuss the target classification problem under the cooperative measurement experiments based on micro-Doppler signatures. However, in the real application, the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high Signal-to-Noise-Ratio (SNR) cannot be guaranteed. In the low SNR cases the micro-motion components are contaminated by the noise and the noised micro-Doppler signatures further affect the parameter estimation or feature extraction in the classification methods. Thus how to improve the noise-robustness of the feature extraction on micro-Doppler signatures is an important research subject to be further studied.

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